MODEL IAA100

Analog Amplifier with Voltage Output



FEATURES

- Bipolar output, differential input
- 256 selectable shunt and gain combinations (DIP Switch)
- Digitally controlled remote shunt with externally accessible activation button
- Sensor polarity reversal DIP switch
- Zero shift DIP switch (Selectable 0 and 5 VDC)
- Reliable spring loaded DIN clip design
- Removable magnetic cover for instrument setting access and protection
- External chassis connection (Refer to Guide)
- Hot swappable with on-board ESD and over voltage/ current protection

MATERIAL & MANUFACTURING

- High-reliability PCB/assembly (IPC CLASS III)
- High-temperature rated material/BOM items
- Strictly controlled inspection, test, and calibration process
- Fully controlled handling/packaging process
- Extensive test and validation for 100% of production units
- Individual calibration report/certification for instruments and systems

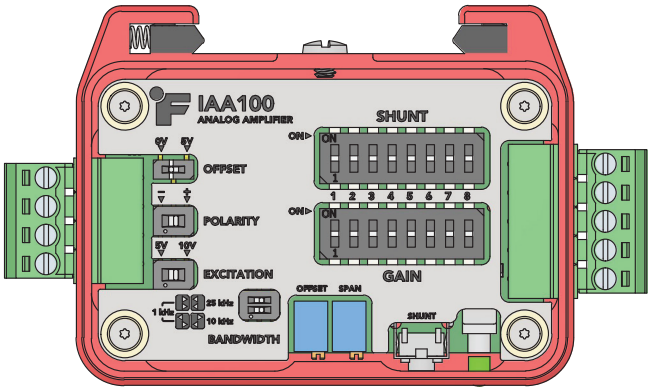
APPLICATIONS

- Medical
- Aerospace
- Precision Industrial Automation
- Lab & Field Instrumentation
- Precision Manufacturing

SPECIFICATIONS	
GENERAL	
Item Number	FSH04461
Material	Aluminum Body/Stainless Steel Cover
Weight	0.23lb [105g]
IP Rating	IP40
INPUT	
Input Type	Strain Gauge (Differential Input from ±0.2 to ±500 mV/V)[4]
Internal Shunt Value	30, 43.7, 60.4, 87.6, 100, 150, 300[3], 432[3] kΩ (±0.1%)
Non-Linearity	±0.005% of FSR[1]
Bridge Excitation	5 or 10 VDC (DIP Switch)
Min/Max Bridge Resistance	87.5 to 5000 Ω
OUTPUT	
Gain Range	110 to 6300[2, 3]
Output Range (single ended)	±5, ±10 VDC
Load Impedance	> 4000 Ω
Bandwidth (Hz)	See Chart[5]
Noise	See Chart[5]
Calibration Span/Offset range	±10% of FSR[1]
POWER	
Supply	12 to 30 VDC
Inrush Current	1.2 A (Max)
Power Consumption	1 W (Max) (instrument only)
Power Indicator	Green LED
ENVIRONMENT	
Operating Temperature	-13° to 158°F [-25° to 70°C]
Storage Temperature	-40° to 185°F [-40° to 85°C]
Temperature Stability/Drift	±10 ppm of FSR[1]/°C
CONFORMITY	
RoHS	2011/65/EU Compliant
CE	FCC 15.107:2022; FCC 15.109:2022; FCC 15.109(g):2022; ICES-003 issue 7:2020; VCCI-CISPR 32:2016; EN IEC 61326-1:2021

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DIP SWITCHES CONFIGURATION

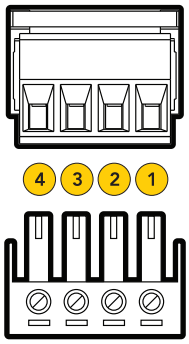


NOISE

Bandwidth kHz	Default Gain mVp-p	Max Gain mVp-p
1	8	15
10	10	25
25	14	35

SENSOR SIDE (item #GOD04252)

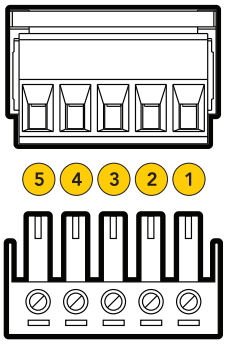
PIN	WIRING CODE	PIN FUNCTIONALITY	COLOR
1	**+ E**	+ Excitation	Red
2	**+ S**	+ Signal	Green
3	**– S**	– Signal	White
4	**–E/GND**	– Excitation	Black



For 6 wire sensors, connect +SENSE to +EXCITATION and –SENSE to –EXCITATION.

POWER/OUTPUT (item #GOD04253)

PIN	WIRING CODE	PIN FUNCTIONALITY	COLOR
1	**+VIN**	Power Supply	Red
2	**GND**	Power Ground	Black
3	**SHUNT**	Remote Connection	Orange
4	**GND**	Output Ground	Blue
5	**VOUT**	Output Signal	Green



General Notes:

[1] FSR = Full Scale Range

[2] Gain = 2 If all switches are in OFF position

[3] ±1% Tolerance

[4] The minimum differential input is determined based on a 10 V excitation and maximum gain to attain the maximum output range

[5] See "Noise" chart located in page 2 for detailed information

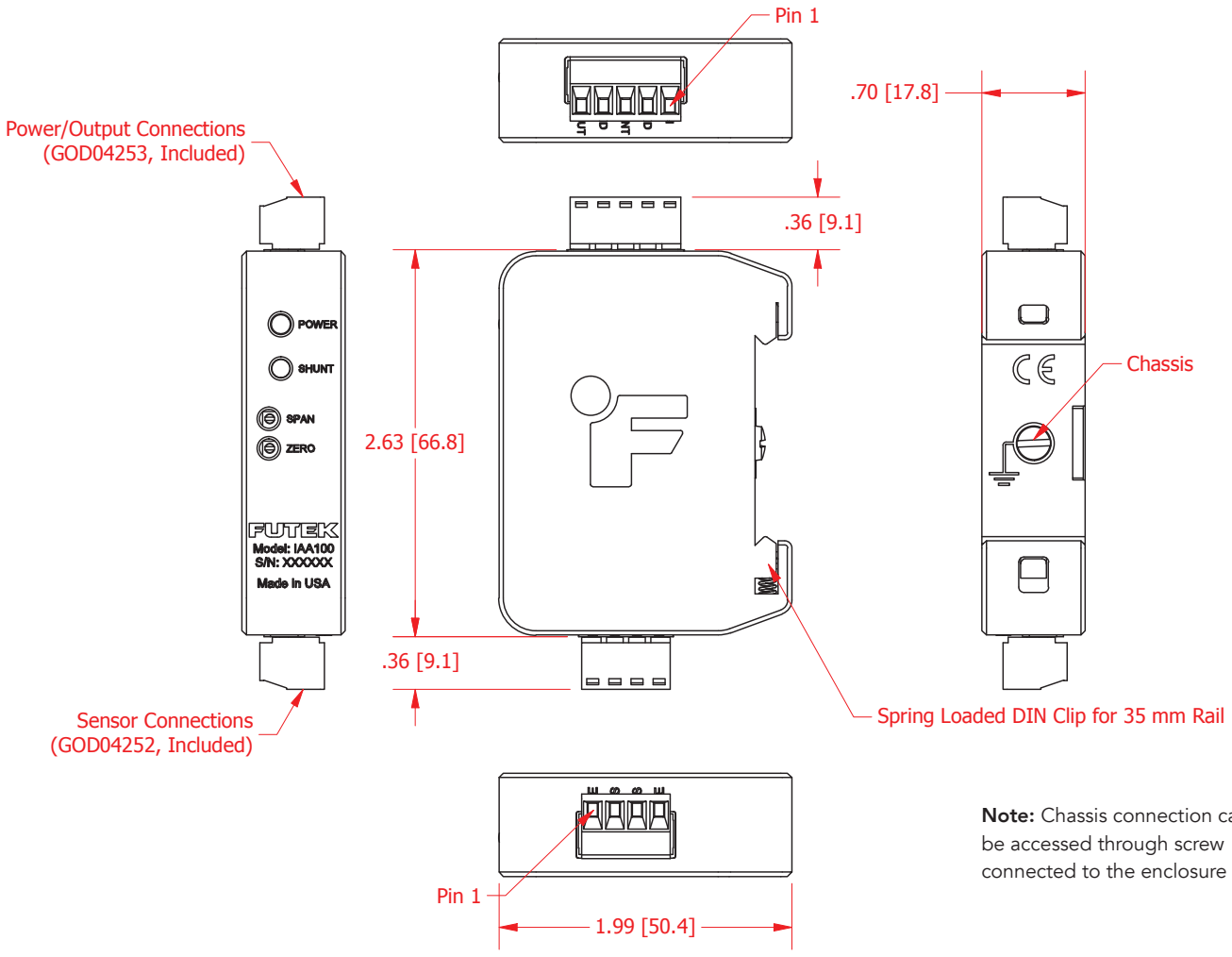
All Grounds are internally connected

Use Chassis for systems with cable Shield

All parameters specified on this drawing have been validated for default settings (2 mV/V and 1 kHz bandwidth)

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DIMENSIONS inches [mm]



Note: Chassis connection can be accessed through screw connected to the enclosure body.

Drawing Number: FI1573

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